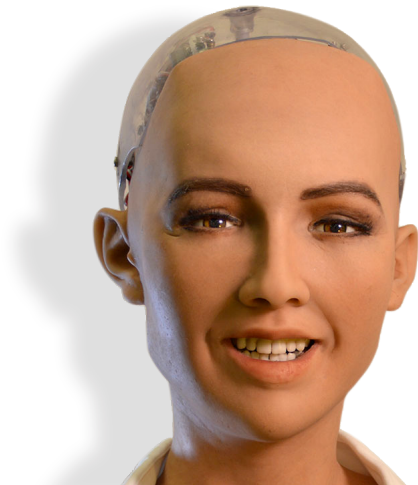


CHAPTER THREE

Artificial Intelligence



♣ Is it Possible to build A Machine intelligent as Human?

- Artificial Intelligence is composed of two words: Artificial and Intelligence
 - Artificial means "man-made," and
 - Intelligence defines "thinking power", or "the ability to learn and solve problems"
- Hence Artificial Intelligence means "a man-made thinking power."
- It is the imitation of human thinking to solve problems that humans cannot.

♣ How Humans think?

- Achieving this goal might require many more years.
- Intelligence or the cognitive process is composed of three main stages:
 - Observe and input the information or data in the brain.
 - Interpret and evaluate the input that is received from the surrounding environment.
 - Make decisions as a reaction towards what you received as input and interpreted and evaluated.

♣ Is it Possible to build A Machine intelligent as Human?

- Artificial Intelligence is composed of two words: Artificial and Intelligence
 - Artificial means "man-made," and
 - Intelligence defines "thinking power", or "the ability to learn and solve problems"
- Hence Artificial Intelligence means "a man-made thinking power."
- It is the imitation of human thinking to solve problems that humans cannot.

♣ How Humans think?

- Achieving this goal might require many more years.
- Intelligence or the cognitive process is composed of three main stages:
 - Observe and input the information or data in the brain.
 - Interpret and evaluate the input that is received from the surrounding environment.
 - Make decisions as a reaction towards what you received as input and interpreted and evaluated.

Mapping Human Thinking — > AI Components

- Because AI is the science of simulating human thinking, it is possible to map the human thinking stages to the layers or components of AI systems.
- In the first stage, humans acquire information from their surrounding environments through human senses such as sight, hearing, smell, taste, and touch, through human organs, such as eyes, ears, and other sensing organs, for example, the hands.

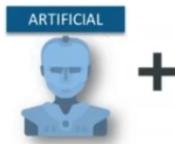
Defination

- Its First Coined by " John MacCarthy"
- The science and engineering of making intelligent machines, especially intelligent computer programs.
- Comp. Programs :- algorithms enabled by constraints.
- is also a theory and development of computer system able to do human tasks which needs human intelligence.
- Such as:
 - Speech recognition
 - translate languages
 - The ability of decision making
 - Speech recognition and other



Defination

- Its First Coined by " John MacCarthy"
- The science and engineering of making intelligent machines, especially intelligent computer programs.
- Comp. Programs :- algorithms enabled by constraints.
- is also a theory and development of computer system able to do human tasks which needs human intelligence.
- Such as:
 - Speech recognition
 - translate languages
 - The ability of decision making
 - Speech recognition and other



Defination

- Its First Coined by " John MacCarthy"
- The science and engineering of making intelligent machines, especially intelligent computer programs.
- Comp. Programs :- algorithms enabled by constraints.
- is also a theory and development of computer system able to do human tasks which needs human intelligence.
- Such as:
 - Speech recognition
 - translate languages
 - The ability of decision making
 - Speech recognition and other



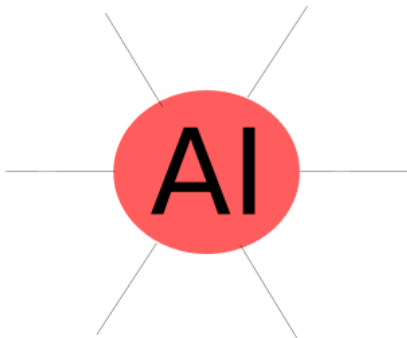
Defination

- Its First Coined by " John MacCarthy"
- The science and engineering of making intelligent machines, especially intelligent computer programs.
- Comp. Programs :- algorithms enabled by constraints.
- is also a theory and development of computer system able to do human tasks which needs human intelligence.
- Such as:
 - Speech recognition
 - translate languages
 - The ability of decision making
 - Speech recognition and other



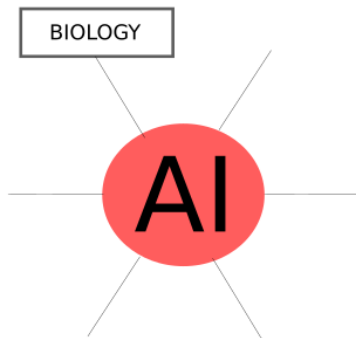
- AI is an interdisciplinary field that requires knowledge of other discipline such as:

Areas which contribute to Artificial Intelligence (AI)



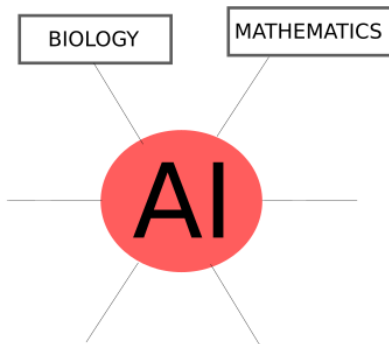
- AI is an interdisciplinary field that requires knowledge of other discipline such as:

Areas which contribute to Artificial Intelligence (AI)



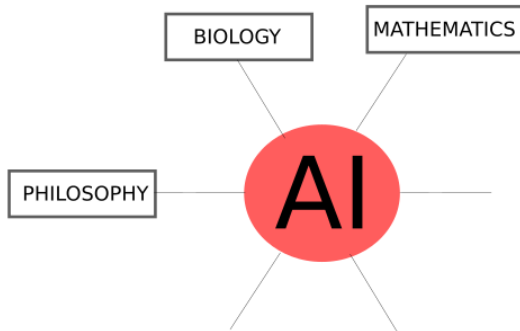
- AI is an interdisciplinary field that requires knowledge of other discipline such as:

Areas which contribute to Artificial Intelligence (AI)



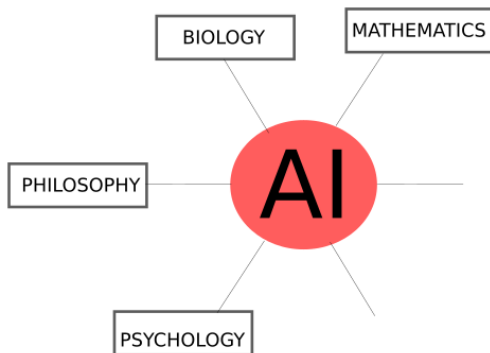
- AI is an interdisciplinary field that requires knowledge of other discipline such as:

Areas which contribute to Artificial Intelligence (AI)



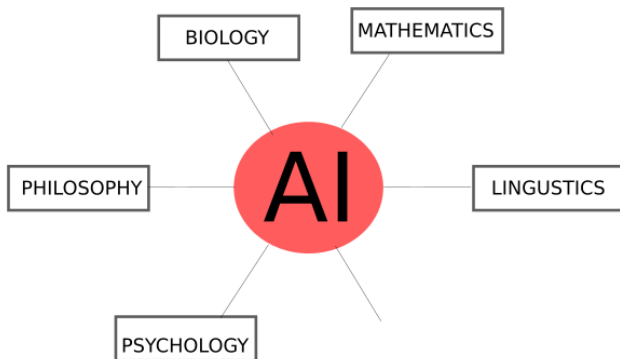
- AI is an interdisciplinary field that requires knowledge of other discipline such as:

Areas which contribute to Artificial Intelligence (AI)



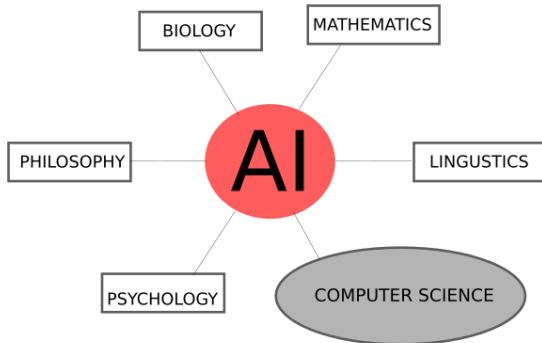
- AI is an interdisciplinary field that requires knowledge of other discipline such as:

Areas which contribute to Artificial Intelligence (AI)



- AI is an interdisciplinary field that requires knowledge of other discipline such as:

Areas which contribute to Artificial Intelligence (AI)



- AI comprises integration of several technologies:
 - Machine learning
 - Natural language processing
 - Reasoning and Perception
- AI deals with the area of developing computing systems that are capable of :
 - Recognizing objects
 - Recognizing and making sense of speech
 - Decision making in a constrained environment.

Artificial Intelligence

- The intelligence should be exhibited by **thinking**, **making decisions**, **solving problems**, more importantly **by learning**.

Machine Learning

- is subset of AI which lets machines to automatically learn from experience and **improve without being explicitly instructed** OR
- is scientific study of algorithms and statistical models that computer systems use to perform a specific task without using explicit instructions, relying on patterns and inference instead
- **Can Machines Really learn, How if ?** **WAIT!**

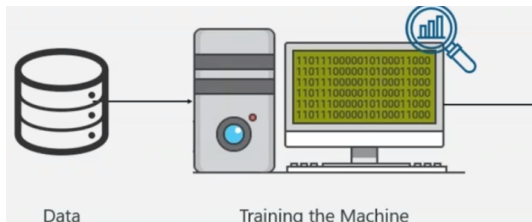


Artificial Intelligence

- The intelligence should be exhibited by **thinking**, **making decisions**, **solving problems**, more importantly **by learning**.

Machine Learning

- is subset of AI which lets machines to automatically learn from experience and **improve without being explicitly instructed** OR
- is scientific study of algorithms and statistical models that computer systems use to perform a specific task without using explicit instructions, relying on patterns and inference instead
- **Can Machines Really learn, How if ?** WAIT!

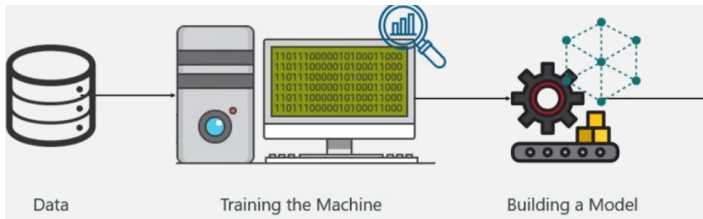


Artificial Intelligence

- The intelligence should be exhibited by **thinking**, **making decisions**, **solving problems**, more importantly **by learning**.

Machine Learning

- is subset of AI which lets machines to automatically learn from experience and **improve without being explicitly instructed** OR
- is scientific study of algorithms and statistical models that computer systems use to perform a specific task without using explicit instructions, relying on patterns and inference instead
- **Can Machines Really learn, How if ?** **WAIT!**

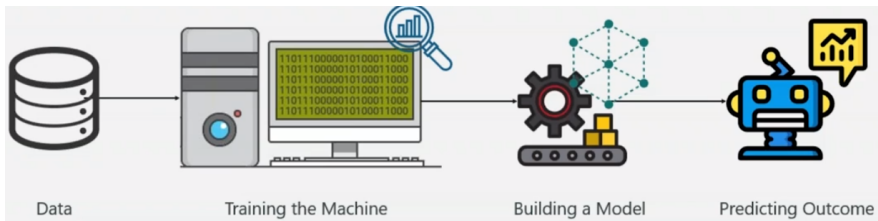


Artificial Intelligence

- The intelligence should be exhibited by **thinking**, **making decisions**, **solving problems**, more importantly **by learning**.

Machine Learning

- is subset of AI which lets machines to automatically learn from experience and **improve without being explicitly instructed** OR
- is scientific study of algorithms and statistical models that computer systems use to perform a specific task without using explicit instructions, relying on patterns and inference instead
- **Can Machines Really learn, How if ? WAIT!**



Terminologies

- ① **Algorithms**: A set of rules and statistical techniques used to learn patterns from data
 - ② **Model**: is Trained data by using ML algorithms.
 - ③
 - ④ **Predictor variable**: is any feature of data that can be used to predict the output.
 - ⑤ **Response variable**: the output variable that needs to be predicted by the predictor variable.
 - ⑥ **Training data**: The machine learning model is built using the training data.
 - ⑦ **Testing data**: is used to evaluate the The ML model
- Does ML has process ? YUP!
 - "Is a process involves building a predictive model that might be used to find out solution for problems"
 - Solution -> Steps ? Wait!

Terminologies

- ① **Algorithms**: A set of rules and statistical techniques used to learn patterns from data
 - ② **Model**: is Trained data by using ML algorithms.
 - ③
 - ④ **Predictor variable**: is any feature of data that can be used to predict the output.
 - ⑤ **Response variable**: the output variable that needs to be predicted by the predictor variable.
 - ⑥ **Training data**: The machine learning model is built using the training data.
 - ⑦ **Testing data**: is used to evaluate the The ML model
- Does ML has process ? YUP!
 - "Is a process involves building a predictive model that might be used to find out solution for problems"
 - Solution -> Steps ? Wait!

Terminologies

- ① **Algorithms**: A set of rules and statistical techniques used to learn patterns from data
 - ② **Model**: is Trained data by using ML algorithms.
 - ③
 - ④ **Predictor variable**: is any feature of data that can be used to predict the output.
 - ⑤ **Response variable**: the output variable that needs to be predicted by the predictor variable.
 - ⑥ **Training data**: The machine learning model is built using the training data.
 - ⑦ **Testing data**: is used to evaluate the The ML model
- Does ML has process ? YUP!
 - "Is a process involves building a predictive model that might be used to find out solution for problems"
 - Solution -> Steps ? Wait!

- Steps to follow

- 1 Defining the purpose/Objective
- 2 Gathering data
- 3 Data preparation
- 4 Data Exploration
- 5 Building a Model
- 6 Evaluating the Model
- 7 Predictions

- What technologies employ AI — > Examples!

- Robotics:

- Deals with the design, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing.
 - These technologies are used to develop machines that can substitute for humans and replicate human actions.

- Machine Automation

- Machine automation is any information technology that is designed to control the work of machines.

- Virtual Reality and Augmented Reality

- Cloud Computing

- "The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer."

- Steps to follow

- 1 Defining the purpose/Objective
- 2 Gathering data
- 3 Data preparation
- 4 Data Exploration
- 5 Building a Model
- 6 Evaluating the Model
- 7 Predictions

- What technologies employ AI — > Examples!

- Robotics:

- Deals with the design, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing.
 - These technologies are used to develop machines that can substitute for humans and replicate human actions.

- Machine Automation

- Machine automation is any information technology that is designed to control the work of machines.

- Virtual Reality and Augmented Reality

- Cloud Computing

- "The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer."

- Steps to follow

- 1 Defining the purpose/Objective
- 2 Gathering data
- 3 Data preparation
- 4 Data Exploration
- 5 Building a Model
- 6 Evaluating the Model
- 7 Predictions

- What technologies employ AI — > Examples!

- Robotics:

- Deals with the design, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing.
 - These technologies are used to develop machines that can substitute for humans and replicate human actions.

- Machine Automation

- Machine automation is any information technology that is designed to control the work of machines.

- Virtual Reality and Augmented Reality

- Cloud Computing

- "The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer."

- Steps to follow

- 1 Defining the purpose/Objective
- 2 Gathering data
- 3 Data preparation
- 4 Data Exploration
- 5 Building a Model
- 6 Evaluating the Model
- 7 Predictions

- What technologies employ AI — > Examples!

- Robotics:

- Deals with the design, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing.
 - These technologies are used to develop machines that can substitute for humans and replicate human actions.

- Machine Automation

- Machine automation is any information technology that is designed to control the work of machines.

- Virtual Reality and Augmented Reality

- Cloud Computing

- "The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer."

- Steps to follow

- 1 Defining the purpose/Objective
- 2 Gathering data
- 3 Data preparation
- 4 Data Exploration
- 5 Building a Model
- 6 Evaluating the Model
- 7 Predictions

- What technologies employ AI — > Examples!

- Robotics:

- Deals with the design, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing.
 - These technologies are used to develop machines that can substitute for humans and replicate human actions.

- Machine Automation

- Machine automation is any information technology that is designed to control the work of machines.

- Virtual Reality and Augmented Reality

- Cloud Computing

- "The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer."

- Steps to follow

- 1 Defining the purpose/Objective
- 2 Gathering data
- 3 Data preparation
- 4 Data Exploration
- 5 Building a Model
- 6 Evaluating the Model
- 7 Predictions

- What technologies employ AI — > Examples!

- Robotics:

- Deals with the design, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing.
 - These technologies are used to develop machines that can substitute for humans and replicate human actions.

- Machine Automation

- Machine automation is any information technology that is designed to control the work of machines.

- Virtual Reality and Augmented Reality

- Cloud Computing

- "The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer."

- Steps to follow

- 1 Defining the purpose/Objective
- 2 Gathering data
- 3 Data preparation
- 4 Data Exploration
- 5 Building a Model
- 6 Evaluating the Model
- 7 Predictions

- What technologies employ AI — > Examples!

- Robotics:

- Deals with the design, construction, operation, and use of robots, as well as computer systems for their control, sensory feedback, and information processing.
 - These technologies are used to develop machines that can substitute for humans and replicate human actions.

- Machine Automation

- Machine automation is any information technology that is designed to control the work of machines.

- Virtual Reality and Augmented Reality

- Cloud Computing

- "The practice of using a network of remote servers hosted on the Internet to store, manage, and process data, rather than a local server or a personal computer."

- Examples ...

- Neural Networks

- is a series of algorithms that endeavors to recognize underlying relationships in a set of data through a process that mimics the way the human brain operates.

- Big Data: Recall it

- Internet of Everything(IoE)

- A system of interrelated **computing devices, mechanical and digital machines, objects, animals or people** that are provided with *unique identifiers* and the ability to transfer data over a network without requiring *human-to-human or human-to-computer interaction*

- Computer Vision/Image Processing

- is an interdisciplinary scientific field that deals with how computers can be made to gain high-level understanding from digital images or videos

- Application areas

- Image recognition, Speech recognition, Chatbots, Natural language generation, and Sentiment analysis.

- Types of Models:

- Deep learning, Machine learning, and Neural Networks.

- Examples ...

- Neural Networks

- is a series of algorithms that endeavors to recognize underlying relationships in a set of data through a process that mimics the way the human brain operates.

- Big Data: Recall it

- Internet of Everything(IoE)

- A system of interrelated **computing devices, mechanical and digital machines, objects, animals or people** that are provided with *unique identifiers* and the ability to transfer data over a network without requiring *human-to-human or human-to-computer interaction*

- Computer Vision/Image Processing

- is an interdisciplinary scientific field that deals with how computers can be made to gain high-level understanding from digital images or videos

- Application areas

- Image recognition, Speech recognition, Chatbots, Natural language generation, and Sentiment analysis.

- Types of Models:

- Deep learning, Machine learning, and Neural Networks.

- Examples ...

- Neural Networks

- is a series of algorithms that endeavors to recognize underlying relationships in a set of data through a process that mimics the way the human brain operates.

- Big Data: Recall it

- Internet of Everything(IoE)

- A system of interrelated **computing devices, mechanical and digital machines, objects, animals or people** that are provided with *unique identifiers* and the ability to transfer data over a network without requiring *human-to-human or human-to-computer interaction*

- Computer Vision/Image Processing

- is an interdisciplinary scientific field that deals with how computers can be made to gain high-level understanding from digital images or videos

- Application areas

- Image recognition, Speech recognition, Chatbots, Natural language generation, and Sentiment analysis.

- Types of Models:

- Deep learning, Machine learning, and Neural Networks.

- Software/Hardware for training and running models:
 - Graphic Processing Units (GPUs),
 - Parallel processing tools (like Spark),
 - Cloud data storage and computer platforms.
- Programming languages for building models
 - **Python**, TensorFlow, Java, and C/C++, etc.

History of Artificial Intelligence

Refer it ...

- Software/Hardware for training and running models:
 - Graphic Processing Units (GPUs),
 - Parallel processing tools (like Spark),
 - Cloud data storage and computer platforms.
- Programming languages for building models
 - **Python**, TensorFlow, Java, and C/C++, etc.

History of Artificial Intelligence

Refer it ...

Types of AI

- Artificial Intelligence can be divided into various types based on :
 - Capability of AI
 - Function of AI

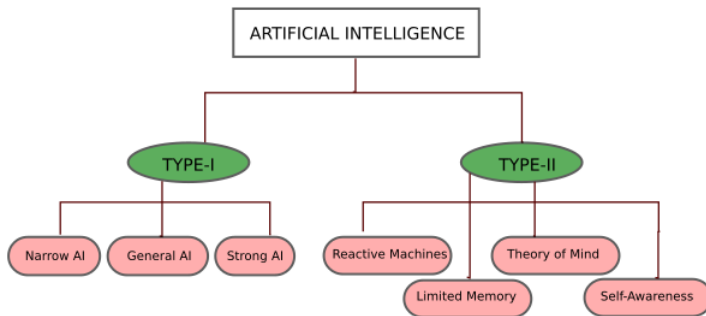


Figure: Classification of AI

Type-I: Based on Capability

• Narrow AI

- is a type of AI which is able to perform a **dedicated task with intelligence**.
- Cannot perform beyond its field or limitations. Coz *as it is only trained for one specific task*
- Hence it is also called **weak AI**
- **Playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition** are some examples of Narrow AI
- **Voice Assistants** i.e Siri(Apple), Cortana (Windows), Alexa, google assistance (only act upon voice commands which perform a certain action)

• General AI

- is type of intelligence **which could perform any intellectual task with efficiency like a human**.
- The idea is to make a system which could be smarter and think like a human by its own.
- General AI systems are still under research

• Super AI(Strong AI)

- **Hard to believe!**
- **Intelligence of Systems at which machines could surpass human intelligence**.
- Can perform any task better than human with cognitive properties.

Type-I: Based on Capability

• Narrow AI

- is a type of AI which is able to perform a **dedicated task with intelligence**.
- Cannot perform beyond its field or limitations. Coz *as it is only trained for one specific task*
- Hence it is also called **weak AI**
- **Playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition** are some examples of Narrow AI
- **Voice Assistants** i.e Siri(Apple), Cortana (Windows), Alexa, google assistance (only act upon voice commands which perform a certain action)

• General AI

- is type of intelligence **which could perform any intellectual task with efficiency like a human**.
- The idea is to make a system which could be smarter and think like a human by its own.
- General AI systems are still under research

• Super AI(Strong AI)

- **Hard to believe!**
- **Intelligence of Systems at which machines could surpass human intelligence.**
- Can perform any task better than human with cognitive properties.

Type-I: Based on Capability

• Narrow AI

- is a type of AI which is able to perform a **dedicated task with intelligence**.
- Cannot perform beyond its field or limitations. Coz *as it is only trained for one specific task*
- Hence it is also called **weak AI**
- **Playing chess, purchasing suggestions on e-commerce site, self-driving cars, speech recognition, and image recognition** are some examples of Narrow AI
- **Voice Assistants** i.e Siri(Apple), Cortana (Windows), Alexa, google assistance (only act upon voice commands which perform a certain action)

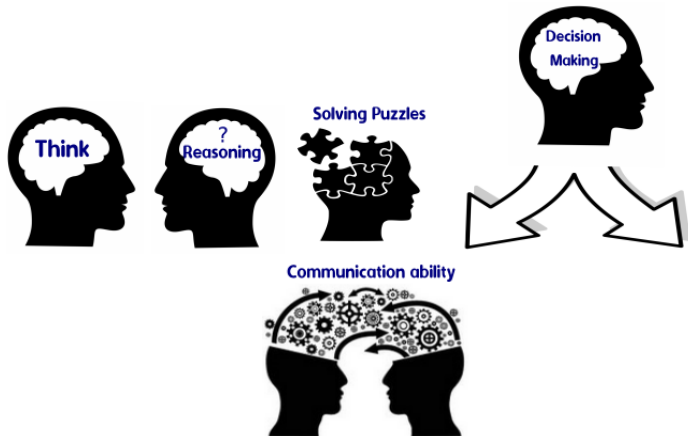
• General AI

- is type of intelligence **which could perform any intellectual task with efficiency like a human**.
- The idea is to make a system which could be smarter and think like a human by its own.
- General AI systems are still under research

• Super AI(Strong AI)

- **Hard to believe!**
- **Intelligence of Systems at which machines could surpass human intelligence**.
- Can perform any task better than human with cognitive properties.

X-stics Strong AI



Type-II AI: Based on functionality

● Reactive Machines

- Such AI systems do not store memories or past experiences for future actions.
- These machines **only focus on current scenarios** and react on it as per possible best action.
- IBM's Deep Blue system, Google's AlphaGo are Examples of Reactive Machine

● Limited Memory

- Machines can store past experiences or some data for a short period of time.
- These machines can use stored data for a limited time period only.
- **Self-driving cars** are one of the best examples of Limited Memory systems.



These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

● Theory of Mind

- AI should understand the human emotions, people, beliefs, and be able to interact socially like humans.
- **Is Such machine existed ?** Check out ..

Type-II AI: Based on functionality

● Reactive Machines

- Such AI systems do not store memories or past experiences for future actions.
- These machines **only focus on current scenarios** and react on it as per possible best action.
- IBM's Deep Blue system, Google's AlphaGo are Examples of Reactive Machine

● Limited Memory

- Machines can store past experiences or some data for a short period of time.
- These machines can use stored data for a limited time period only.
- **Self-driving cars** are one of the best examples of Limited Memory systems.



These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

● Theory of Mind

- AI should understand the human emotions, people, beliefs, and be able to interact socially like humans.
- **Is Such machine existed ?** Check out ..

Type-II of AI

- Self-awareness AI

- Self-awareness AI is the future of Artificial Intelligence.
- These machines will be super intelligent, and will have their own consciousness, sentiments, and self-awareness.
- These machines will be smarter than human mind.
- Self-Awareness AI does not exist in reality still and it is a hypothetical concept.

- Artificial Intelligence has various applications in today's society.
- Currently, AI is being applied across several industries.



"But we cannot say that AI is replacing humans but it is certainly making the work of human beings more efficient."

- So where Does AI applied. Wait!

Artificial Intelligence.....



Figure: Application Areas of AI

1 AI in Agriculture

- Now a day's agriculture is becoming digital
- Agriculture is employing AI to to yield healthier crops, control pests, monitor soil and growing conditions, organize data for farmers, help with workload,
- Moreover, improve a wide range of agriculture-related tasks in the entire food supply chain.
- Agricultural Robot's
 - Autonomous robots to handle essential agricultural tasks such as harvesting crops at a higher volume and faster pace than human laborers.
 - Moreover, AI is used:
 - solid and crop monitoring,
 - predictive analysis
 - Harvesting and more ..

② AI in Health

- AI applications are revolutionizing how the **health sector works to reduce spending and improve patient outcomes.**
- The primary aim of health-related AI applications is *to analyze relationships between prevention or treatment techniques and patient outcomes.*
- **IBM Watson and Chatbots**⇒ healthcare technologies

② AI in education

- AI can automate grading so that the tutor can have more time to teach.
- AI chatbot can communicate with students as a teaching assistant.
- AI in the future can be work as a personal virtual tutor for students, which will be accessible easily at any time and any place.

② AI in e-commerce

- AI is providing a competitive edge to the e-commerce industry,
- it is becoming more demanding in the e-commerce business.
- AI is helping shoppers to discover associated products with recommended size, color, or even brand.

② AI in Automotive Industry

- Some Automotive industries are using AI to provide virtual assistant to their user for better performance.
- Such as Tesla has introduced TeslaBot, an intelligent virtual assistant.
- Various Industries are currently working for developing self-driven cars which can make your journey more safe and secure.

⑥ AI in Data Security

- Cyber-attacks are growing very rapidly in the digital world. **Security of data is crucial**
- AI is begin Implemented as means for security.
 - Ex: **AEG bot**, **AI2 Platform**,are used to determine software bug and cyber-attacks in a better way.

⑥ AI in Transportation

⑥ AI in Robotics

⑥ AI in Data Security

- Cyber-attacks are growing very rapidly in the digital world. [Security of data is crucial](#)
- AI is begin Implemented as means for security.
 - Ex: [AEG bot](#), [AI2 Platform](#),are used to determine software bug and cyber-attacks in a better way.

⑥ AI in Transportation

⑥ AI in Robotics

Finished!!!

